Memory and Focus: Illusions of Plausibility

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The Focus Benefit

Linguistic Focus signals relevance and alternatives. In (1), the cleft emphasises that what shattered was *the plate*, not something else:

(1) It was ***the plate*** that shattered

Notably, focus enhances memory for affected constituents:

- Better recall and recognition [1]
- Facilitated error and change detection [2, 3]
- Generation and maintenance of alternatives [4]

Focus and Illusions of Plausibility

[5] found that verbs with implausible target objects were read more quickly if there was a plausible distractor object, as in example (2):

(2) Sue remembered the {plate / letter} (Target) that the butler with a {cup / napkin} (Distractor) accidentally *shattered* in the dining room yesterday.

Problem: Illusions of Plausibility persist when the target is focused! [5]

(3) What Sue remembered was the ***letter*** next to the cup that...

However, pseudo-clefts don't unambiguously signal where focus is.

(4) **What** did Sue remember?
What Sue remembered was [***the plate*** *next to the napkin*]...

(5) **Which plate** did Sue remember?
What Sue remembered was [the plate *next to* [***the napkin***]]...

Question: When controlling for focus location and contrastive alternatives [6], can focus on the target obviate Illusions of Plausibility?

Experiment Design

Target Focus Context:

Context: The household staff generally try to clear up the table before attempting any other chores. The butler was in a rush this morning, though, and bumped into *a tray of dishes* while he was tidying up. When the footman walked in, he noticed a few shards of porcelain on the carpet.

Question: What was it that the butler had broken?

Distractor Focus Context:

Context: The household staff generally try to clear up the table before attempting any other chores. The butler was in a rush this morning, though, and bumped into a tray of dishes while he was tidying up. He dropped a letter which had been left by *a dirty mug*, and put it back next to the ones still on the table.

Question: Which letter did the butler drop?

Experimental Sentence:

Item: # It was the letter next to the cup that the butler shattered...

Sensibility Judgment:

Judgment: Was that a good answer to the question? (*Yes/No*)

The Current Study

Separate Maze-task [7, 8] (n=62) and eye-tracking while reading (n=44) experiments manipulated target plausibility (±*plaus*), distractor plausibility (±*plaus*), and focus position (*target* vs *distractor*). Trials consisted of a *context*, followed by the *experimental sentence*, and a *sensibility judgment*.

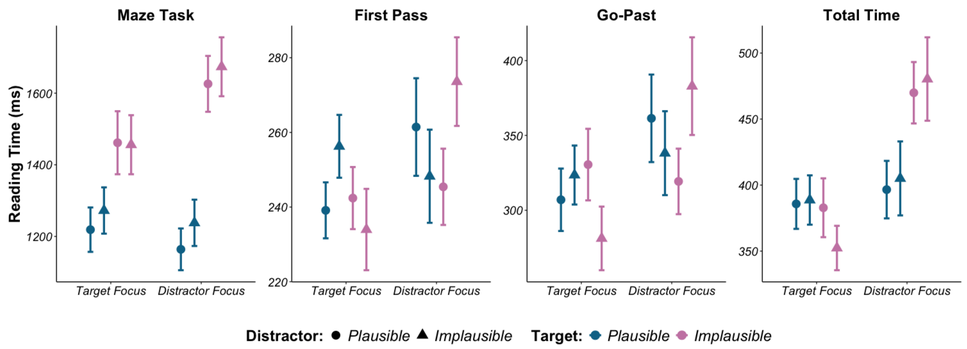

Figure 1. *Mean by-participant response (Maze) and reading (ET) times at the critical verb (ms). Error bars are standard error.*

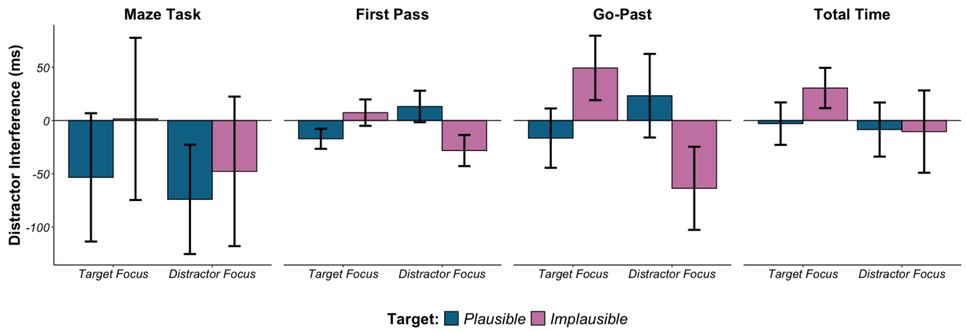

Figure 2. *Mean by-participant "distractor interference" effects (plausible distractor — implausible distractor) at the critical verb (ms). Negative values indicate plausible-distractor facilitation. Error bars are standard error.*

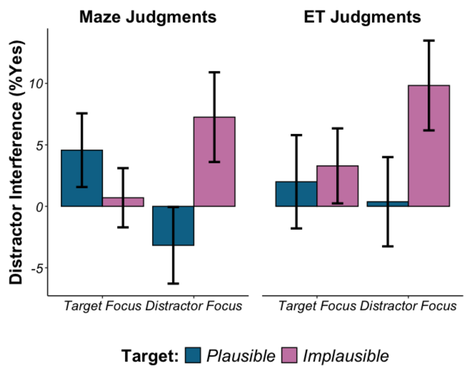

Figure 3. *Mean by-participant "distractor effects" (plausible — implausible) proportion of "yes" responses in post-sentence judgments. Positive values indicate plausible-distractor facilitation. Error bars are standard error.*

Primary Results

- At the verb, target focus led to faster reading overall, and reduced the penalty for target-implausibility
- A significant target×distractor×focus interaction in first pass ($\hat{\beta}$=-7.6, t=2.4) and go-past ($\hat{\beta}$=-21.5, t=2.2) times

→ Plausible distractors facilitated reading for implausible sentences, but only in distractor-focus contexts

- A significant target×distractor×focus interaction in sensibility judgments in the maze-task ($\hat{\beta}$=1.05, p<0.05)

→ Plausible distractors improved the sensibility of implausible sentences, but only in distractor-focus contexts

Discussion

- Illusions of Plausibility: We replicate the primary finding in [5], showing that verbs with implausible objects are read faster, and accepted more often, if there is a plausible "distractor" in the sentence
- Illusions and Focus: Unlike [5], we find that focus *does* affect plausibility illusions

→ No evidence of a plausibility illusion when the *target* was focused

→ Reliable evidence of a plausibility illusion when the *distractor* was focused

- Focus and Reading: Focus on the target led to *faster* reading at the verb, and reduced the penalty of implausibility in both experiments. We think this reflects focus directing attention in reading

→ Targets which were *infelicitous* answers to the context question elicited longer reading times

→ This may have reduced the implausibility penalty at the verb because readers *already knew* the sentence was implausible

Conclusions

This study produced three primary findings:

(*i*) Focus directs attention during reading: focus facilitated implausibility detection when on the target, but induced misinterpretation when on the distractor (the illusion of plausibility)

(*ii*) Focus modulates illusions of plausibility: the illusion of plausibility effects are found only when focus is on the *distractor*

(*iii*) Illusions of plausibility arise early and can have lasting effects: the plausibility illusion effect emerges as early as first pass reading, and the illusion can linger, as seen in the judgment task

References

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